



Town of Crystal

Solar Energy Ordinance

Section 1. Title

This Ordinance shall be known and may be cited as the “Town of Crystal Solar Energy Ordinance”.

Section 2. Purpose

The purpose of this ordinance is to establish a municipal review procedure and performance standards for Solar Energy Systems (SES), including those typically characterized as “solar farms”. These standards are intended to:

- a. Establish clear guidelines, standards and time frames for the Town to regulate Solar Energy Systems.
- b. Permit the Town to fairly and responsibly protect public health, safety and welfare.
- c. Minimize any potential adverse effect of solar development on surrounding land use.
- d. Provide for the decommissioning/removal of panels and associated utility structures that are no longer being used for energy generation and transmission purposes.
- e. To conserve the environment, wildlife habitat and unique natural areas.
- f. To promote the general welfare of the community.

Section 3. Applicability

This Ordinance shall apply to Ground Mounted Solar Energy Systems applications filed with the Planning Board

Section 4. Definitions

Solar Energy System (SES); a solar photovoltaic cell, module, array, or solar hot air or water collector device, including all Solar related Equipment, which relies upon solar radiation as an energy source for collection, inversion, storage and distribution of solar energy for electricity generation or transfer of stored heat.

Solar Energy System - Ground Mounted; A Solar Energy System that is structurally mounted to the ground and is not roof-mounted; may be of any size (small, medium or largescale). For purposes of this Ordinance the Ground Mounted Solar Energy System: has a physical size no greater than 5 acres that is not directly connected to a residential structure.

Solar Energy System, Roof Mounted; A Solar Energy System that is mounted on the roof of a building or structure; may be of any size (small, medium or large-scale).

Solar Energy; Radiant Energy (direct, diffuse and/or reflective) received from the sun.

Solar Array; A grouping of multiple solar modules with the purpose of harvesting solar energy.

Table 1.

Scale of SES	Ground-space definitions	Power definitions	Application fee	Permit fee
Commercial Solar Energy Systems (CSES)	≥ 3,000 square feet < 2 acres (87,120 square feet)	≥ 31 kW < 500 kW	\$500	\$1.00 per kW
<u>Industrial Solar Energy Systems (ISES)</u>	≥ 2 acres (87,120 square feet) but less than 5 acres	≥ 500 kW	\$2,500	\$1.00 per kW
<u>Private Residential Solar Energy Systems (PRSES)</u>	< 2,000 square feet (additional terms apply)	< 21 kW	Subject to terms of the Crystal Building Code	No permit fee required
<u>Residential-Business Solar Energy System (RBSES)</u>	< 3,000 square feet	< 31 kW	\$150	\$1.00 per kW with a min. of \$25

Section 5. Dimensional Requirements

- A. Height: A Ground or Pole Mounted Solar Energy System shall have a maximum height of 15 feet as measured from the ground level to the system's highest point at full tilt.
- B. Minimum front setback shall be 1,000 feet, minimum side setback shall be 1,000 feet and minimum back setback shall be 1,000 feet.
- C. Lot coverage for Ground Mounted Solar Energy Systems shall be calculated based on the area of the supporting structure that is in contact with the ground and not the area of panels as long as the area under the panels remains vegetated.
- D. Add cover membrane over land so chemicals can't seep into ground.

Section 6. General Standards for all Arrays

- 1) Array placement must be designed to minimize or negate any solar glare onto nearby properties or roadways.
- 2) If lighting is provided at the site, lighting shall be shielded and downcast such that the light does not spill onto the adjacent parcel or the night sky. Motion sensor control is preferred.
- 3) Any point of potential contact of people or animals with generated electric current must be secured.

Section 7. Exempt Arrays

The following are exempt from this Ordinance:

- 1) Roof-mounted on any legal permitted residential or residential accessory structure.
- 2) Ground-or pole mounted for private use, with a panel area less than 3,000 square feet.
- 3) Building integrated solar (i.e., shingle, hanging solar, canopy, etc...).
- 4) Repair or replacement of array components that do not enlarge the area of the existing array.

Section 8. Solar Array Complex Plan Review

All non-exempt arrays must be approved by the Crystal Planning Board through this Ordinance.

The following requirements must be included in a Solar Energy Conversion Array application:

- 1) A site plan showing property lines, the location of any wetlands or flood zones, the location of proposed panels, equipment, fencing and access roads, and the location and setback of any roads or streets.
- 2) A decommissioning plan signed by the party responsible for decommissioning and the landowner (if different) whose minimum requirements meet the standards in section 5 of this Ordinance. Such a plan must be filed in the Aroostook County Registry of Deeds prior to the first operation of the array.
- 3) A Waste Stream Management Plan (WSMP) for the construction waste and debris at the site of the said array, including but not limited to cardboard, wood, scrap metal, scrap wire and clearing and grading wastes from the construction site and the disposal sites) of such waste. Information on the amount of material that is being recycled shall be included in the WSMP. The Code Enforcement Officer shall conduct a final inspection to ensure compliance with the approved plan.

Section 9. Guarantee For Removal

At the time of approval of a proposed array, and prior to initiating construction of any array within the Town of Crystal, the applicant must guarantee the costs for the removal of the facility.

- 1) The amount of guarantee shall be equal to 125% of the estimated removal cost, provided by the applicant and certified by a professional civil engineer licensed in Maine or a professional array construction company.
- 2) The owner of the facility shall provide the Planning Board with a revised removal cost estimate and structural evaluation prepared by a professional civil engineer licensed in Maine or a professional array construction company every (50 years from the date of the Planning Board's approval of the Solar Array Complex Plan.
- 3) Types and Contents of Guarantee – One of the following performance guarantees chosen by the applicant shall be provided on approval of the application.
 - a. **Interest-Bearing Escrow Account-** A cash contribution equal to 125% of the estimated removal cost for the establishment of an escrow account shall be made by either a certified check made out to the Town, direct deposit into a savings account, or purchase of a certificate of deposit.
 - i. For any account opened by the applicant, the Town shall be named as owner or co-owner, and consent of the Town shall be required for withdrawal.
 - ii. Any interest earned on the escrow shall be returned to the applicant unless the Town has found it necessary to draw on the account, in which case the interest earned shall be proportionately divided between the amount returned to the applicant and the amount withdrawn to complete the required work.
 - b. **Performance Bond** – A performance bond shall detail the conditions of the bond, the method of release of the entire bond or portions of the bond to the Town, and the

procedures for collection by the municipality. The bond documents shall specifically reference the array facility for which approval is sought.

- c. **Irrevocable Letter of Credit** – An Irrevocable Letter of Credit from a bank or other lending institution shall indicate that funds have been set aside for the removal of the array facility and may not be used for any other project or loan. The letter of credit shall detail the procedures for collection by the municipality. The conditions and amount of the performance guarantee shall be determined by the Planning Board with the advice of the Town Selectmen, and/or Town Attorney, expenses paid for by the applicant.

Section 10. Screening, Security, Maintenance and Regulatory Compliance

- A. The lots on which Ground Mounted Solar Energy Systems are located shall be buffered from roads and residences by plantings, berms and natural topographical features.
- B. Subject to the approval of the Crystal Planning Board, the lots on which Ground Mounted Solar Energy Systems are located shall be protected by perimeter fence. One or more signs shall be affixed to the fence identifying the owner of the facility and emergency contact information.
- C. For purposes of emergency services, the owner or operator of a Ground Mounted Solar Energy Systems shall provide a copy of the project summary, electrical schematic, and site plan to the emergency services that cover the Town of Crystal. Upon request, the owner or operator shall cooperate with the Fire Department in the developing of an Emergency Response Plan. All means of shutting down the system shall be clearly marked on the plan. The owner or operator shall identify a responsible person for the Code Enforcement Officer and the Fire Chief for public inquiries throughout the life of the installation. The town shall not be responsible for any fires involving the energy system.
- D. The owner or operator of a Ground Mounted Solar Energy System shall maintain the facility in good condition. Maintenance shall include but not be limited to, painting structural repairs, vegetation control and integrity of security measures. Site access shall be maintained to a level acceptable to the Crystal Fire Chief. The owner or operator shall be responsible for the cost of maintaining the access road(s) unless the road(s) is accepted as a public way.
- E. The owner or operator of a Ground Mounted Solar Energy System shall build and maintain it in compliance with all relevant Federal, State and Local Laws, Regulations and Ordinances.

Section 11. Decommissioning and Removal

- A. Any Ground Mounted Solar Energy System that has reached the end of its useful life, ceases to generate power or has been abandoned shall be removed pursuant to a plan approved by the Planning Board during the application process. The owner or operator shall remove physically the installation no more than 180 days after the date of discontinued operations. The owner or operator shall notify the Code Enforcement Officer

- by certified mail, return receipt requested, of the proposed date of the discontinued operations and plans for removal.
- B. Decommissioning shall consist of: (1) physical removal of all solar energy systems, structures, equipment, security barriers and transmission lines from the site; (2) disposal of all solid and hazardous waste in accordance with Local, State and federal waste disposal regulations; and (3) stabilize or re-vegetation of the site as necessary to minimize erosion. The Code Enforcement Officer may allow the owner or operator to leave landscaping or designated below-grade foundations to minimize erosion and disruptions to vegetation.
 - C. Absent a notice of proposed date of decommissioning or written notice of extenuating circumstances, Ground Mounted Solar Energy Systems shall be considered abandoned when it fails to generate electricity for more than one year without having first obtained the written consent of the Code Enforcement Officer. Determination of abandonment shall be made by the Code Enforcement Officer.
 - D. If the owner or operator of a Ground Mounted Solar Energy System fails to remove the installation in accordance with the requirements of this section within 180 days of abandonment or the proposed date of decommissioning, the Town of Crystal retains the right to use the performance guarantee and any and all legal or available means necessary to cause an abandoned, hazardous or decommissioned solar energy system to be removed.

Section 12. Decommissioning and Abandonment

- 1) The owner or operator of the facility, or the owner of the parcel if there is no separate owner or operator of the facility or if the owner/operator fails to do so, shall do the following as a minimum to decommission the project:
 - a. Remove all non-utility owned equipment, conduits, structures, fencing and foundations to a depth of at least four feet below grade.
 - b. Revegetate any cleared areas with appropriate plantings that are native to the region according to an approved Solar Array Complex plan, unless requested in writing by the owner of the real estate to not revegetate due to plans for agriculture planting or other development subject to the Planning Board's approval.
 - c. Fill in all holes, depressions shall occur within 12 months of the facility ceasing to operate.
- 2) All said removal and decommissioning shall occur within 12 months of the facility ceasing to operate.
- 3) Abandonment will occur because of any of the following conditions unless the lessee or owner of the facility or of the parcel notifies the Code Enforcement Officer of the intent to maintain and reinstate the operation of the facility within 30 days of the following events:
 - a. The land lease (if applicable) ends; or
 - b. The system does not function for 12 months; or
 - c. The system is damaged and will not be repaired or replaced.

In the event the lessee of the facility fails to decommission the facility as outlined above, the landowner shall decommission the facility within 90 days of notice by the Code Enforcement Officer.

Section 13. Authority

To the extent that any provision of this Ordinance is deemed invalid by a court of competent jurisdiction, such provisions shall be removed from the Ordinance and the balance of the Ordinance shall remain valid.

1) In the event the Planning Board requires expert opinions, advice, or testimony, during the course of reviewing the application to determine the impact to surrounding properties or public safety implications, or to resolve any other issues regarding the proposal, it shall first use due diligence to obtain and utilize free services from governmental or non- profit sources.

2) Should the Planning Board be unable to obtain and utilize free services, the Board may authorize the hiring of independent third-party consultants to review array proposals to determine the impact to surrounding properties or public safety implications or resolve any other issues regarding the proposal. The Planning Board shall require the applicant to pay for such services after giving notice to the applicant of the name of the expert, the area of qualification of the expert, and the purpose for which the expert is required and the approximate cost of the expert.

3) The applicant shall be provided with an opportunity to meet with the Code Enforcement Officer to arrange a schedule for payment costs.

4) The applicant shall have the right to request a public hearing before the Appeals Board to determine if the experts, as noticed by the Planning Board, are necessary to a determination of any issue properly before the Planning Board, and if the approximate cost of the expert are reasonable. It will be the applicant's burden to prove that the requested expert is unnecessary or that the cost is not reasonable.

Section 14. Specific Application Requirements

An application for a Solar Energy System Permit must include the following, at the cost of the applicant:

1) A description of the owner of the SES, the operator if different, and detail of qualifications and track record to run the facility.

2) If the operator will be leasing the land, a copy of the agreement (minus financial compensation) clearly outlining the relationship inclusive of the rights and responsibilities of the operator, landowner and any other responsible party, with regard to the SES and the life of the agreement.

3) A construction plan and timeline, identifying known contractors, site control and anticipated on-line date

4) An operations and maintenance plan, including site control and the projected operating life of the system; such a plan shall include measures for maintaining safe access to the installation, stormwater controls, as well as general procedures for operational maintenance of the installation. Additionally, such plans shall include efforts to promote beneficial flora and fauna (ex. honeybees, butterflies, etc.) as well as a commitment to not using pest-control substances (ex. Pesticides, herbicides, fungicides, and/or insecticides).

5) An emergency management plan for all anticipated hazards.

6) A description of the work required to physically remove all Solar Energy System and Solar Related Components, including associated foundations, buildings, cabling, electrical components, and any other associated facilities to the extent they are not otherwise in or proposed to be placed

into productive use. All earth disturbed during decommissioning must be graded and re-seeded, unless the landowner of the affected land requests otherwise in writing and subject to the Planning Board approval.

Glare- All SES shall be situated to eliminate concentrated glare onto nearby structures or roadways.

Lighting- Lighting shall be limited to that required for safety and operational purposes and shall be shielded from interference with abutting properties.

Section 15. Application Fees

- 1) Industrial Solar Energy Systems (ISES). The Application Fee is \$2,500.
- 2) Commercial Solar Energy Systems (CSES). The Application Fee is \$500.
- 3) Residential Business Solar Energy Systems (RBSES). The Application Fee is \$150.
- 4) Private Residential Solar Energy systems (PRSES). For the purpose of assessing application fees for Private Residential Solar Energy Systems, the SES shall be considered an Accessory Structure or use as defined in the Town of Crystal Building Code Ordinance, and any application fee shall be assessed accordingly. The Application Fee will be a minimum of \$25.

Section 16. Effective Date and Duration

This Ordinance shall take effect upon enactment by the Town of Crystal unless otherwise provided and shall remain in effect until it is amended or repealed.

Section 17. Enforcement Violations and Penalties

This Ordinance shall be enforced by the municipal officers or their designee. Violation of this Ordinance shall be subject to the enforcement and penalty provisions of Title 30-A, M.R.S.A. 4452, Enforcement of Land Use Laws and Ordinances.

Adopted by voters at the Special Town Meeting, 07/06/2023

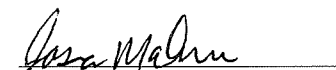
Select Board Crystal,


Patrick York


Calvin White

Phyllis Chambers

Gerald Violette


Jason MacArthur